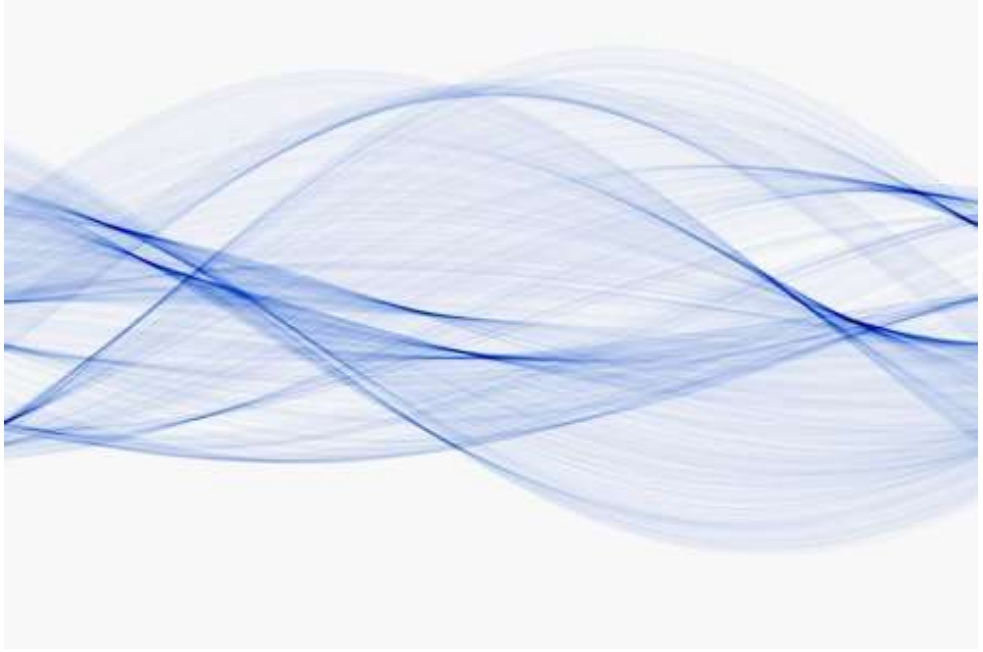
Wave Natures and Ray Transmission

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The Nature of Light

There are three theories explain the nature of light:

- Quantum Theory – Light consists of small particles (photons). This theory better explains light detection and generation processes.
- Wave Theory – Light travels as a *transverse* electromagnetic wave.
- Ray Theory – Light travels along a straight line and obeys laws of *geometrical optics* – useful tool when the objects are much larger than the wavelength of light

The dual nature of light

Scientists have observed that light energy can behave like a wave as it moves through space, or it can behave like a discrete particle with a discrete amount of energy (quantum) that can be absorbed and emitted. As we study and use light, both models are helpful. Two models are:

- ✓ Quantum Theory of Light
- ✓ Wave Theory of Light

Wave Nature of Light



• **Newton (1680) b**elieved in the particle theory of light. In **reflection** and **refraction**, light behaved as a particle. He explained the straight-line casting of sharp shadows of objects placed in a light beam. But he could not explain the textures of shadows

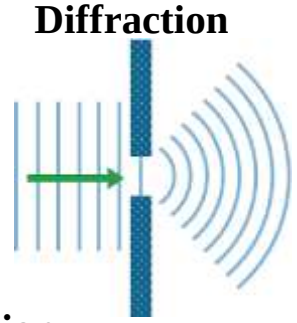


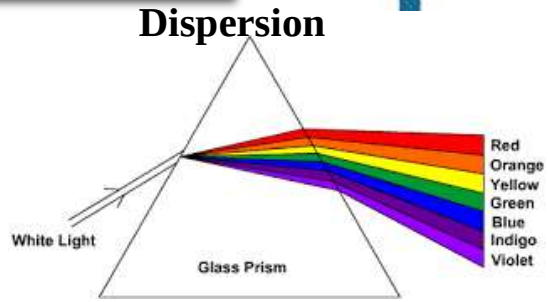
• **Young (1800) –** Showed that light interfered with itself. **Wave theory:** Explains interference, diffraction, and dispersion.

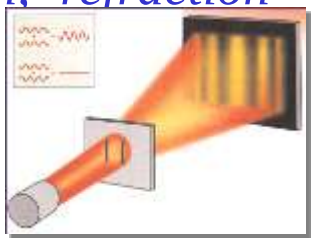
•The wave theory is also able to account for the fact that the edges of a shadow are not quite sharp.

This theory describes: *Propagation, reflection, refraction* and *attenuation*









G Ekspong, Stockholm University, Sweden, 1999.

Wave Nature of Light - *contd.*



James Clerk Maxwell (1850) - Mathematical theory of electromagnetism led to the view that light is of electromagnetic nature, propagating as a wave from the source to the receiver.



Heinrich Hertz (1887) - Discovered experimentally the existence of electromagnetic waves at radio-frequencies.

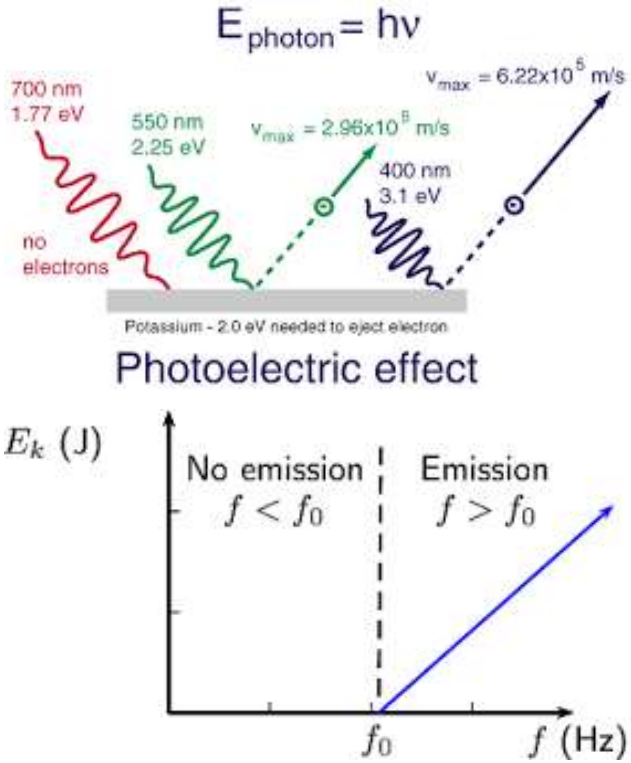
Wave theory does not describe the absorption of light by a photosensitive materials

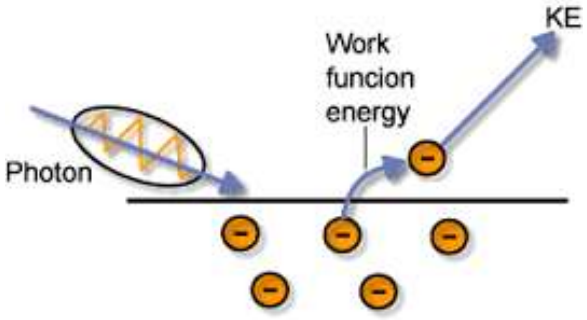
He discovered the photoelectric effect. So what is it?

Wave Nature of Light - *contd.*

Light (electromagnetic wave) shining on a metal surface results in its energy being transferred to the electrons on the surface of the metal and some electron will escape. The released electron are called "**photo-electron**".

$E_{\text{photo-electron}}$ is related to the f_{incident} light but **not its intensity**. So light is energetic enough to shift electron(s) or not.





The work function – The amount of E need by EM wave to free an electron from a meta surface (i.e., a $E_{\text{threshold}}$).

1900-20 Max Planck, Neils Bohr and Albert Einstein

Invoked the idea of light being emitted in tiny pulses of energy

Wave Nature of Light - *contd.*

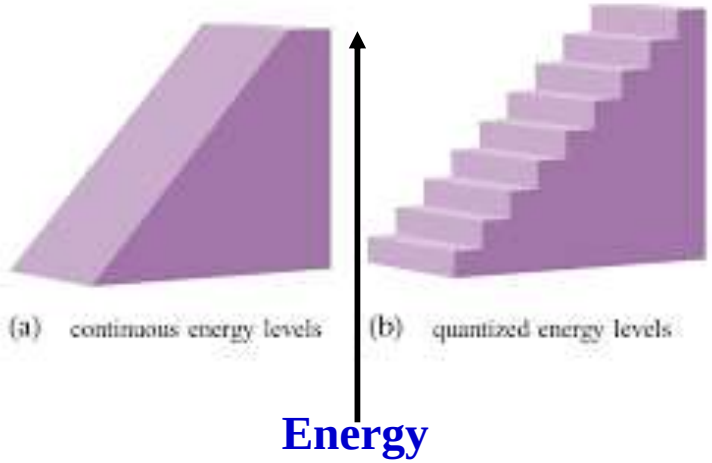


Planck (1900) - Developed a model that explained light as a quantization of energy [Got a Nobel Prize]. Considered the Block Body cavity -

E = nhf *n = 0, 1, 2, 3.., h = Plancks constant 6.626 x 10^{-34} JS [or 4.136 x 10^{-15} eVs] , f = frequency of one of the standing wave within a black body.*

- The energy spectrum of the standing wave within the black body cavity is not continuous as in the classical theory, but take specific values:

Classical **Quantum**



E.g., - A standing wave within a black body cavity has a frequency f = 7.25 x 10^{14} Hz (Blue-violet).
The energy is:
E = n(4.136 x 10^{-15} eVs)(7.25 x 10^{14} Hz) = n(3 eV),

So for n = 1, 2, 3, 4, .. we have E = 3, 6, 9, 12, …. But not 1, 2, 4, 5, 7, 8 and so on.

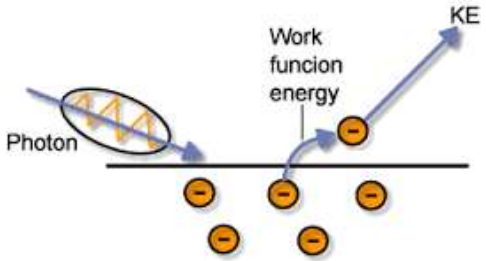
Interesting!

This contradicted the continuous wave theory of light, where wave DO NOT have distinct constituent!

Wave Nature of Light - *contd.*



Einstein (1905) – Used Plank's idea to showed that in the photoelectric effect (light causing electrons to be emitted from a metal surface) **light must act as a particle.**



He came up with this

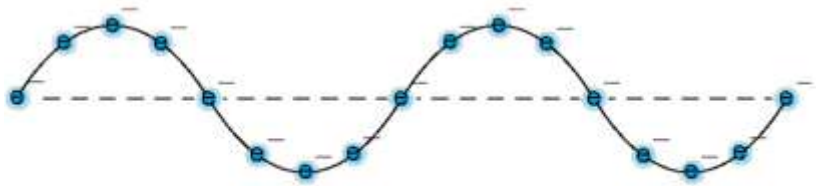
$$K_{max} = hf - \phi$$

Max. kenetic Energy of an electron

Work function

Therefore, light must be regarded as having a dual nature;

- **in some cases light acts as a wave**
- **in others it acts like a particle.**



Wave Theory of Light

- Electromagnetic light signal has electric and magnetic fields orthogonal to each other.
- The frequency of this EM wave is in the order of THz. Therefore, it is convenient to measure it in terms of wavelength.

$$c = \nu\lambda$$

- where, *c* - speed of light 3 X 10^8 m/s in air, ν - frequency and λ- wavelength
- Ex: Find the ν when λ = *1550 nm.*
- *Answer: 193.5 THz | v=c/ λ=3*10^8/1550*10^-9*

Particle Nature of Light

Light behaviour can be explained in terms of the amount of energy imparted in an interaction with some other medium. In this case, a beam of light is composed of a stream of small lumps or QUANTA of energy, known as PHOTONS. Each photon carries with it a precisely defined amount of energy defined as:

$$W_p = h*f \qquad \textbf{Joules (J)}$$

where; h = Plank's constant = 6.626×10^{-34} J.s, f = Frequency Hz

The convenient unit of energy is electron volt (eV), which is the kinetic energy acquired by an electron when accelerated to 1 eV = 1.6×10^{-19} J.

- Even although a photon can be thought of as a **particle of energy** it still has a fundamental wavelength, which is equivalent to that of the propagating wave as described by the wave model.
- This model of light is useful when the light source contains only a few photons.

Particle Theory Vs. Wave Theory

Newton's particle theory explain Only four properties of light:

- ✓ rectilinear propagation,
- ✓ reflection,
- ✓ refraction,
- ✓ and dispersion.

Wave theory (Huygens' wave theory.) of light explains most phenomena involving light:

- ✓ propagation in straight line
- ✓ reflection
- ✓ refraction
- ✓ superposition, interference, diffraction
- ✓ polarization
- ✓ Doppler effect

Light: Particle or a Wave?

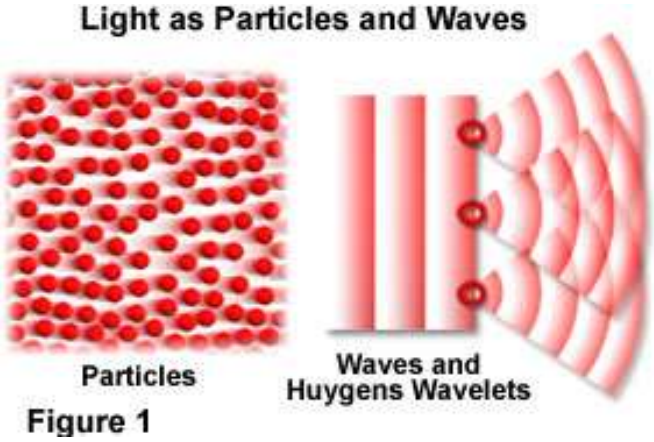


Figure 1

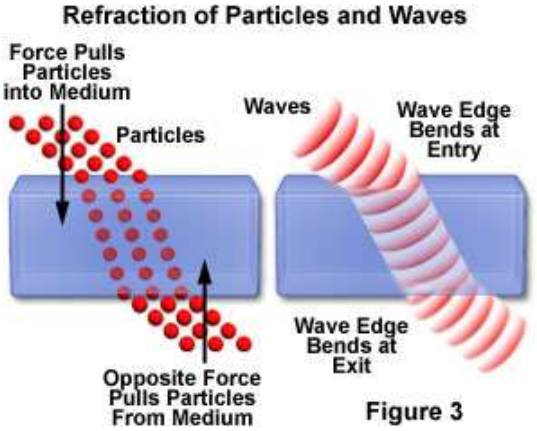


Figure 3

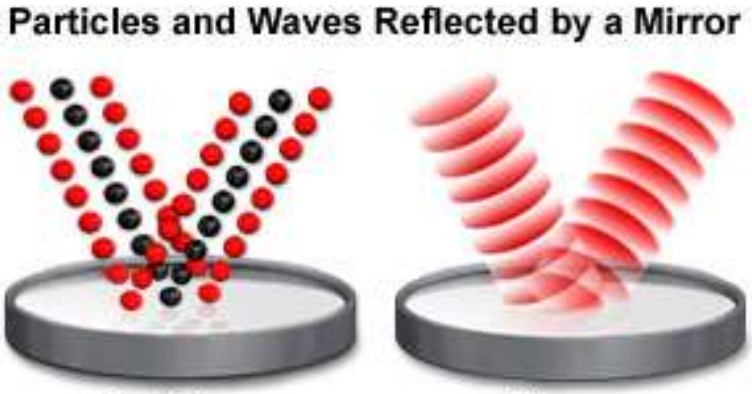


Figure 4

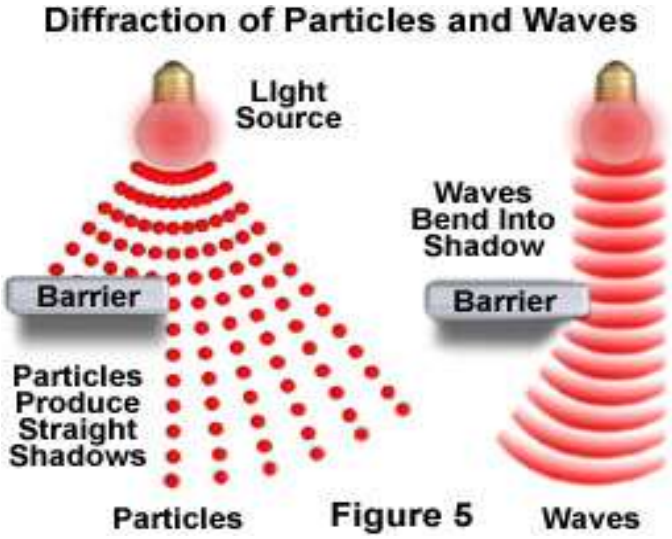


Figure 5

Light: Particle or a Wave?(continue)

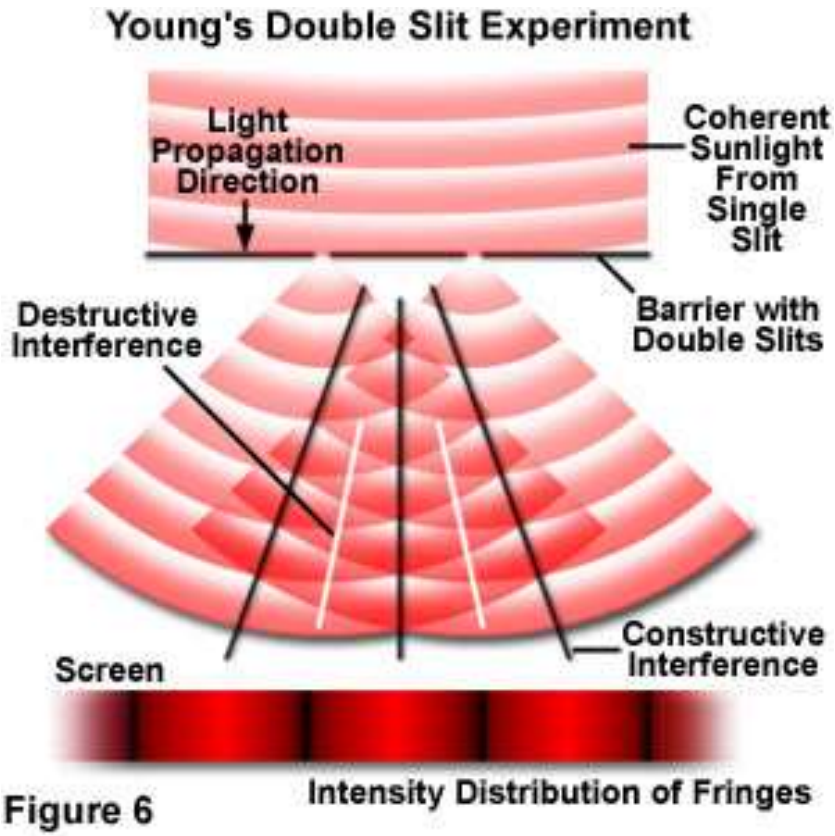


Figure 6

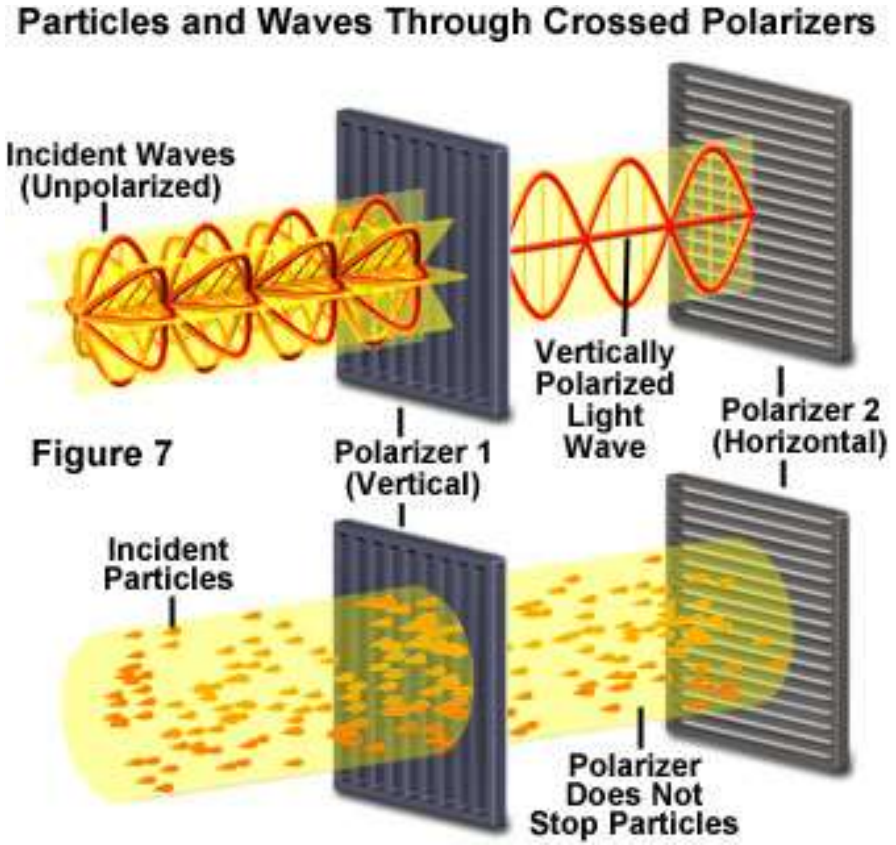


Figure 7

Electromagnetic Radiation

- Carries energy through space (includes visible light, dental x-rays, radio waves, heat radiation from a fireplace)
- The wave is composed of a combination of mutually perpendicular electric and magnetic fields the direction of propagation of the wave is at right angles to both field directions, this is known as an

ELECTROMAGNETIC WAVE

EM wave move through a vacuum at 3.0 x 10^8 m/s ("speed of light")

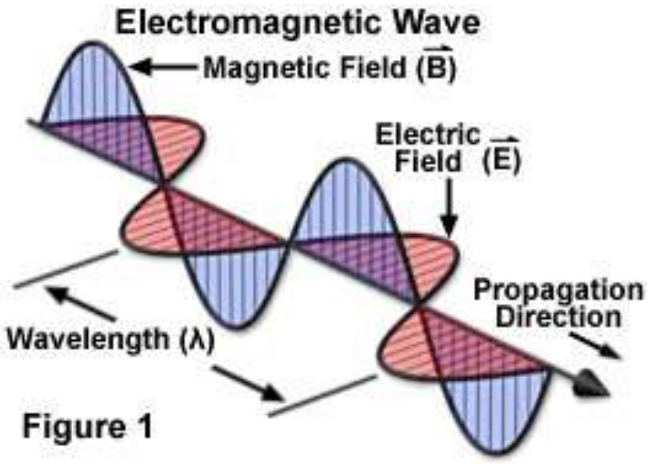
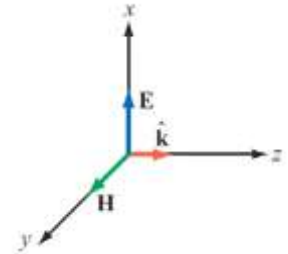
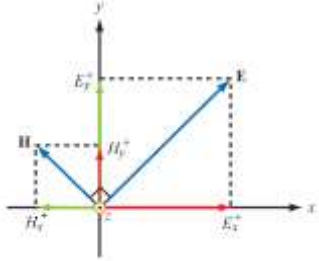


Figure 1





$$\boldsymbol{E} = \boldsymbol{E}_0(z,\phi)e^{j(\omega t-\beta z)}$$

$$\boldsymbol{H} = \boldsymbol{H}_0(z,\phi)e^{j(\omega t-\beta z)}$$

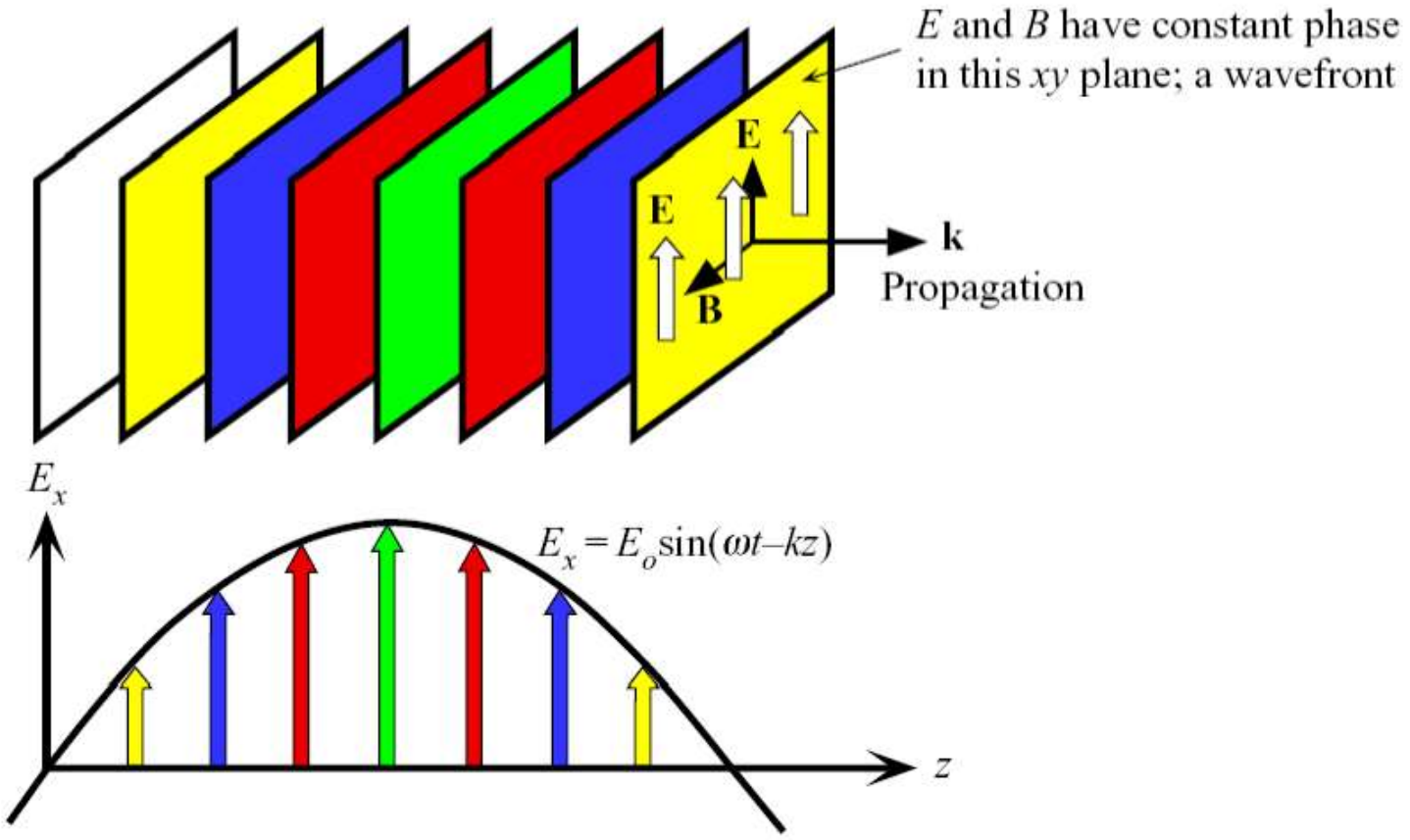
$$E = v_p H = H\,[c/n]$$

Speed of light in a vacuum $c = f \times \lambda_o$

β - Propagation constant = ω/v_p

The **optical field** generally refers to the electric field.

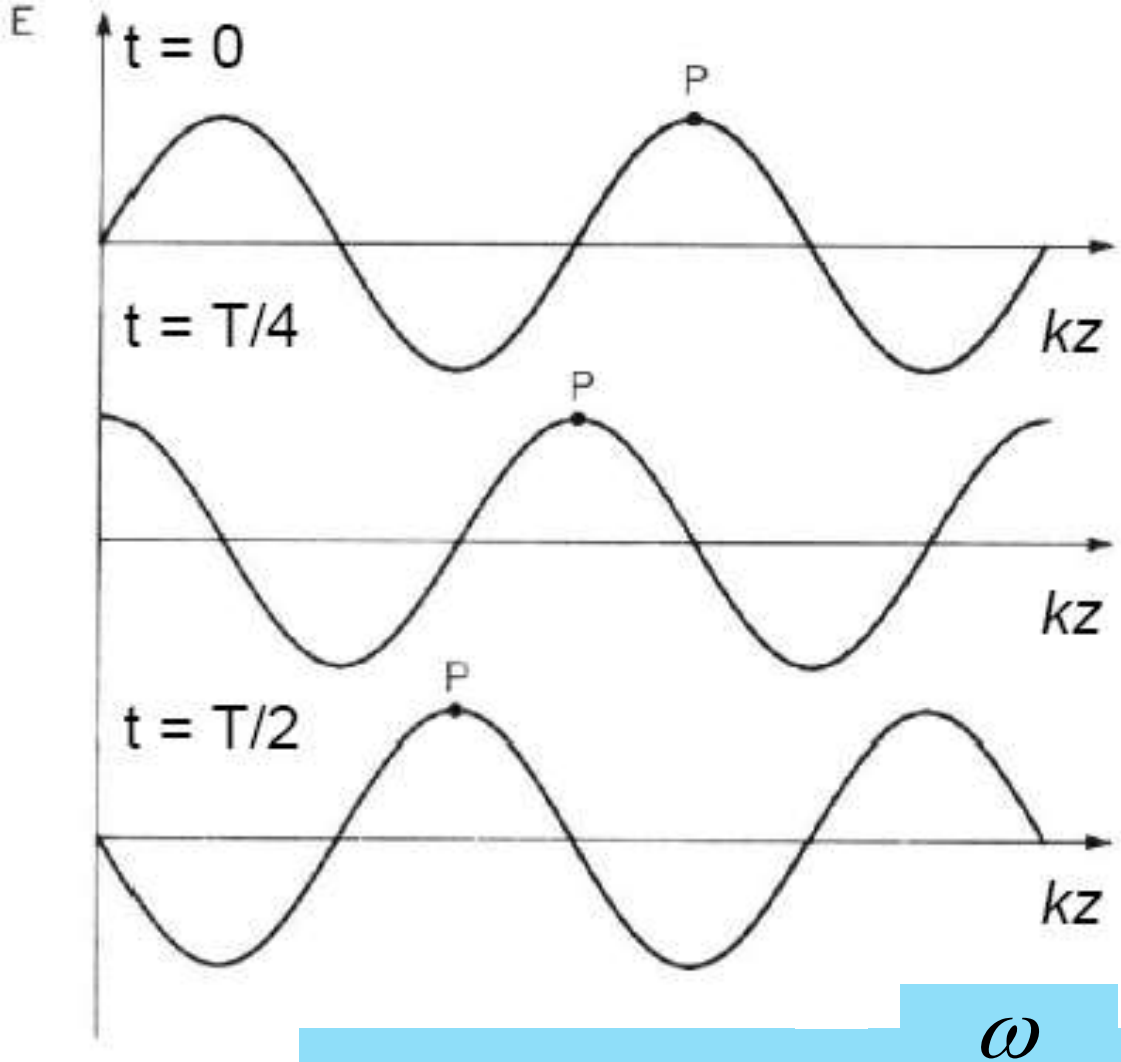
Plane Electromagnetic Wave



A surface over which the phase of a wave is constant is referred to as a **wavefront**.

Phase Velocity

The **phase velocity** of a wave is the rate at which the wave propagates in any medium.



P is a point of **constant phase**

$$\Rightarrow \omega t - kz = \text{constant}$$

$$\Rightarrow \frac{dz}{dt} = \frac{\omega}{k}$$

$$v_p = \frac{\omega}{k}$$ phase velocity

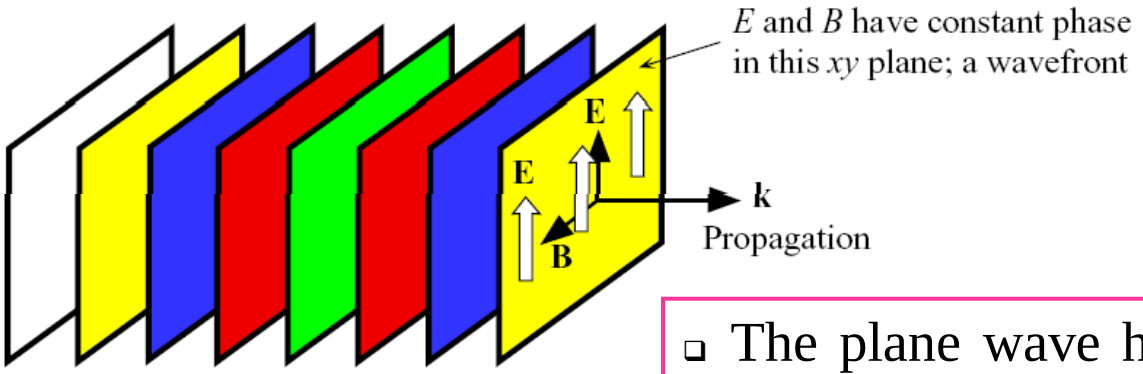
$$c_0 = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 2.998 \times 10^8 \frac{m}{s}$$

Phase velocity: $v_p = \frac{\omega}{k} = \frac{c}{n}$

c_o: Speed of light in air

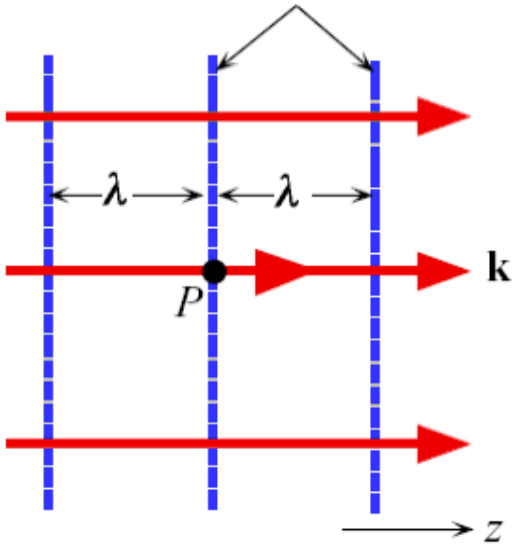
n: Refractive index

Plane Waves



$$E_x = E_0 \cos(\omega t - kz + \phi_0)$$

Wave fronts
(constant phase surfaces)



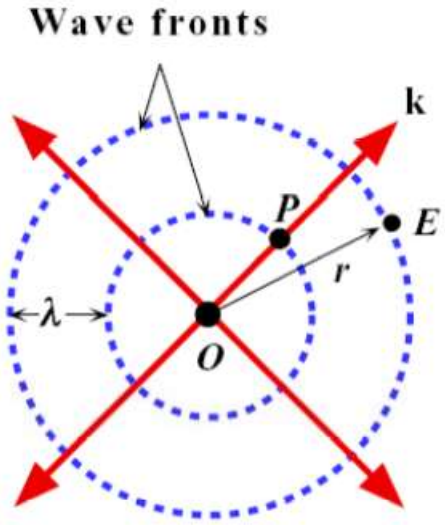
A perfect plane wave

- The plane wave has no angular separation (**no divergence**) in its wavevectors.
- E_0 does not depend on the distance from a reference point, and it is the same at all points on a given plane perpendicular to **k**.
- The plane wave is an idealization that is useful in analyzing many wave phenomena.
- Real light beams would have finite cross sections.
- The plane wave obeys **Maxwell's EM wave equations**.

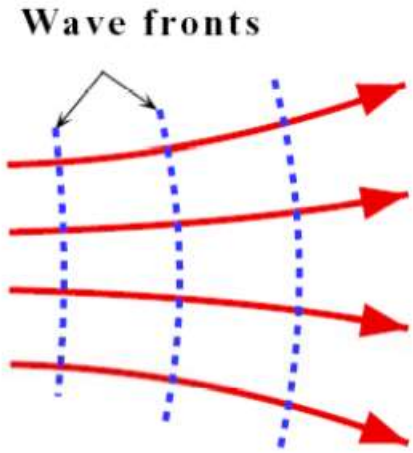
Diverging Waves

$$\frac{\partial^2 E}{\partial x^2}+\frac{\partial^2 E}{\partial y^2}+\frac{\partial^2 E}{\partial z^2}=\varepsilon_0 \varepsilon_r \mu_0 \frac{\partial^2 E}{\partial t^2}$$

$$E = (A/r)\cos(\omega t - kr)$$



A perfect spherical wave

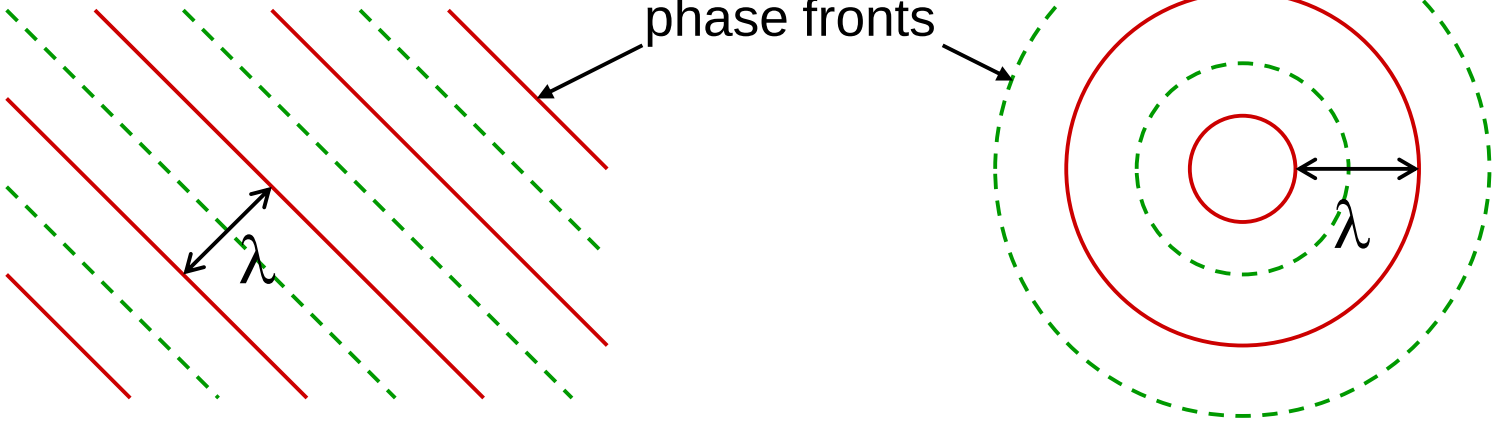


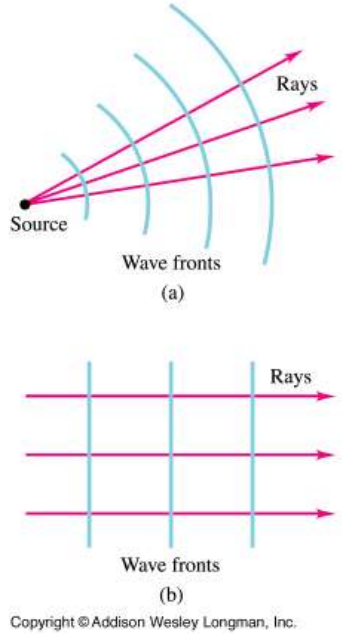
A divergent beam

- A **spherical wave** is described by a traveling wave that emerges from a point EM source and whose amplitude decays with the distance *r* from the source.
- **Optical divergence** refers to the angular separation of wavevectors on a given wavefront.
- Plane and spherical waves represent two extremes of wave propagation behavior from perfectly parallel to fully diverging wavevectors. They are produced by two extreme sizes of EM wave sources: an infinitely large source for the plane wave and a point source for the spherical wave. A real EM source would have a finite size and finite power.

The Wave Equation

Solutions to Maxwell's equations:





Plane wave: $E \propto e^{-jk\cdot r}$

Spherical wave: $E \propto \dfrac{e^{-jk\cdot R}}{R}$

Wave number in vacuum

$$k = \frac{2\pi}{\lambda}$$

$$k = n \cdot k_0$$

$$\lambda = \lambda_0 / n$$

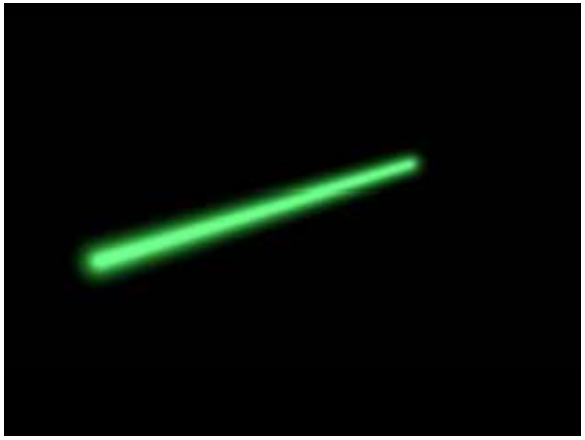
$$k = \omega\sqrt{\varepsilon\mu_0} = \sqrt{\varepsilon_r} \cdot \omega\sqrt{\varepsilon_0\mu_0} = \sqrt{\varepsilon_r} \cdot k_0$$

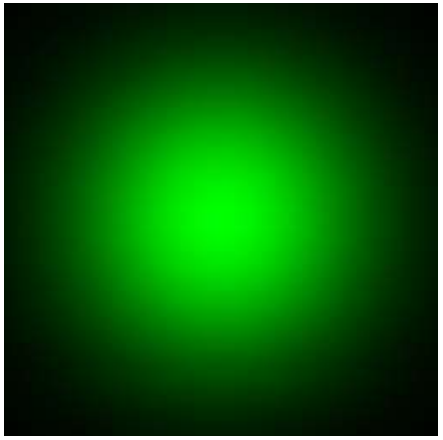
$$n = \sqrt{\varepsilon_r}$$

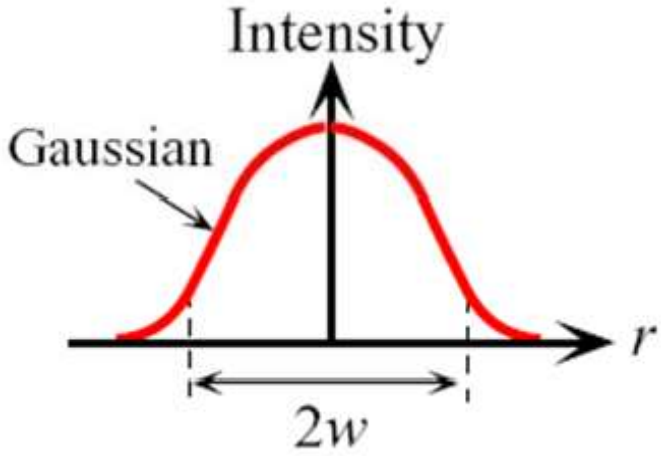
Note: $k = \beta$

Gaussian Beams

Many laser beams can be described by Gaussian beams.

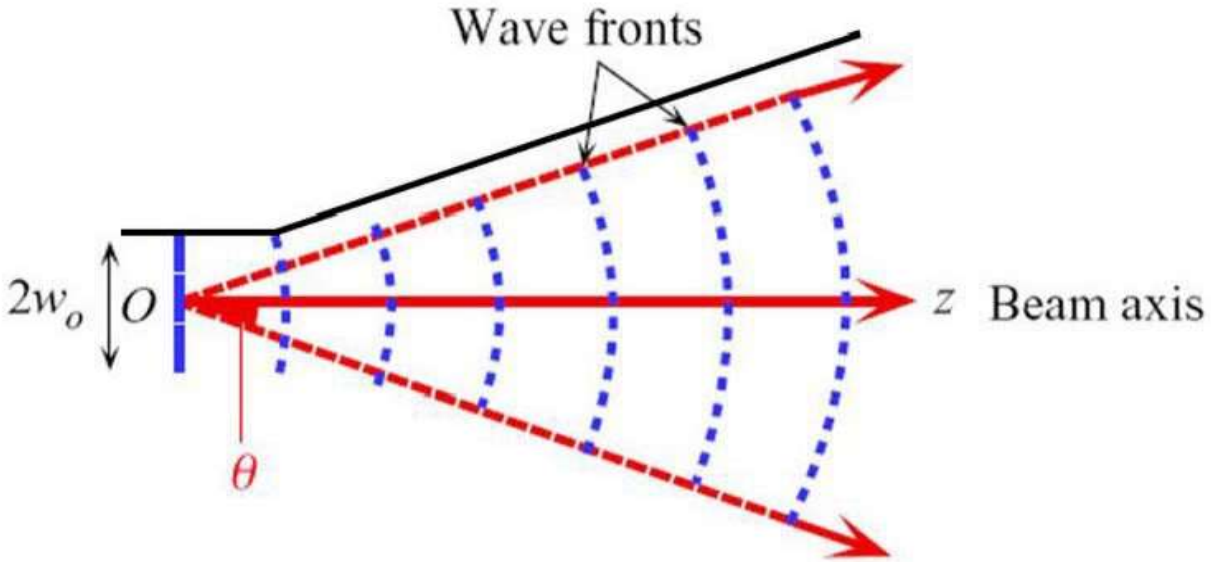






A Gaussian beam has an exp[$j(\omega t - kz)$] dependence to describe propagation characteristics but the amplitude varies spatially away from the beam axis. The beam diameter $2w$ at any point z is defined as the diameter at which the beam intensity has fallen to $1/e^2$ (13.5%) of its maximum.

Gaussian Beams



Diffraction causes light waves to spread transversely as they propagate, and it is therefore impossible to have a perfectly collimated beam. The spreading of a laser beam is in precise accord with the predictions of pure diffraction theory. Even if a Gaussian laser beam wavefront were made perfectly flat at some plane, it would quickly acquire curvature and begin spreading out. The finite width $2w_0$ where the wavefronts are parallel is called the **waist** of the beam. w_0 is the **waist radius** and $2w_0$ is the **spot size**. The increase in beam diameter $2w$ with z makes an angle of 2θ, which is called the **beam divergence**. The spot size and beam divergence are two important parameters when choosing lasers.

Refractive Index

A refractive index of a material defines that how fast light travels through the material.

In vacuum:

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

The phase velocity in a medium is:

$$v = \frac{1}{\sqrt{\varepsilon_0 \varepsilon_r \mu_0}}$$

The refractive index of the medium is:

$$n = \frac{c}{v} = \sqrt{\varepsilon_r}$$

Material	Index of refraction
Air	1.0
Carbon dioxide	1.0
Water	1.33
Ethyl alcohol	1.36
Magnesium fluoride	1.38
Fused silica	1.46
Polymethyl methacrylate	1.49
Silica glass	1.5
Sodium chloride	1.54
Polystyrene	1.59
Calcite	1.6
Silicon	3.5
Gallium arsenide	3.35
Indium gallium arsenide	3.6

Group Velocity

Since there are no perfect monochromatic waves in practice, we have to consider the way in which a group of waves differing slightly in wavelength travel along the same direction. For two harmonic waves of frequencies $\omega - \delta\omega$ and $\omega + \delta\omega$ and wavevectors $k - \delta k$ and $k + \delta k$ interfering with each other and traveling along the z direction:

$$E_{x,1}(z,t) = E_0 \cos\left[(\omega - \delta\omega)t - (k - \delta k)z\right]$$

$$E_{x,2}(z,t) = E_0 \cos\left[(\omega + \delta\omega)t - (k + \delta k)z\right]$$

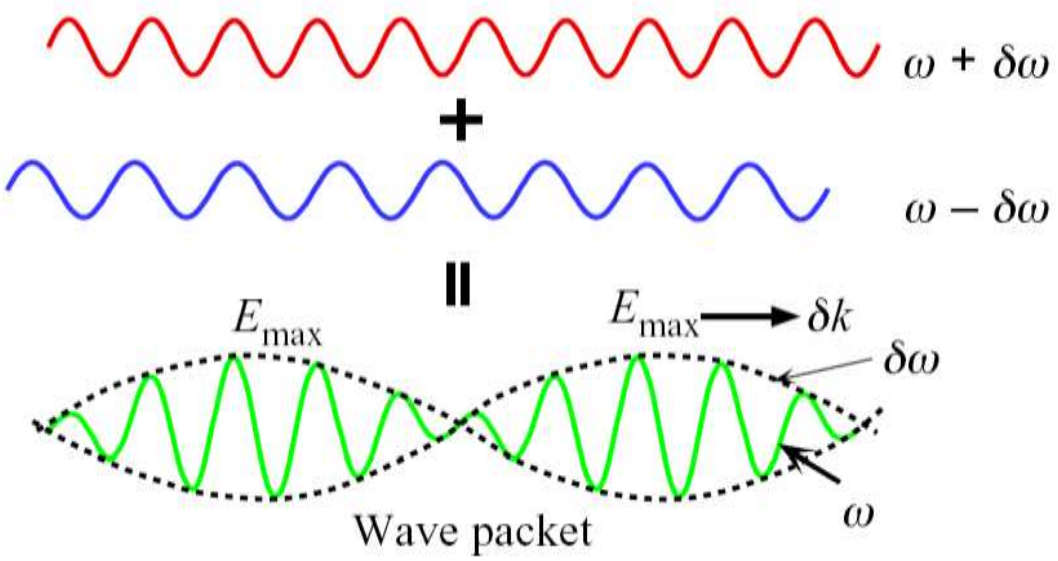
Using

$$CosA + CosB = 2Cos\left[\frac{1}{2}(A - B)\right]Cos\left[\frac{1}{2}(A + B)\right]$$

Group Velocity

We obtain:

$$E_x = E_{x,1} + E_{x,2} = 2E_0 \cos\left[(\delta\omega)t - (\delta k)z\right]\cos(\omega t - kz)$$



Generate a wave packet containing an oscillating field at the mean frequency ω with the amplitude modulated by a slowly varying field of frequency δ.

Group Velocity

$$E_x = 2E_0 \cos\left[(\delta\omega)t - (\delta k)z\right]\cos(\omega t - kz)$$

The maximum amplitude moves with a wavevector δk and a frequency $\delta\omega$.

Group velocity $v_g = \dfrac{d\omega}{dk}$

Phase velocity $v = \dfrac{\omega}{k} = \dfrac{c}{n}$

In vacuum:

$\omega = ck$

$v_g = \dfrac{d\omega}{dk} = c$

group velocity || **phase velocity**

The group velocity defines the speed with which energy or information is propagated since it defines the speed of the envelope of the amplitude variation.

In a medium with n as a function of λ_0:

$$v_g = \frac{d\omega}{dk} = \frac{c}{n - \lambda_0 \dfrac{dn}{d\lambda_0}} = \frac{c}{N_g} \qquad N_g = n - \lambda_0 \frac{dn}{d\lambda_0}$$

N_g is the group index of the medium.

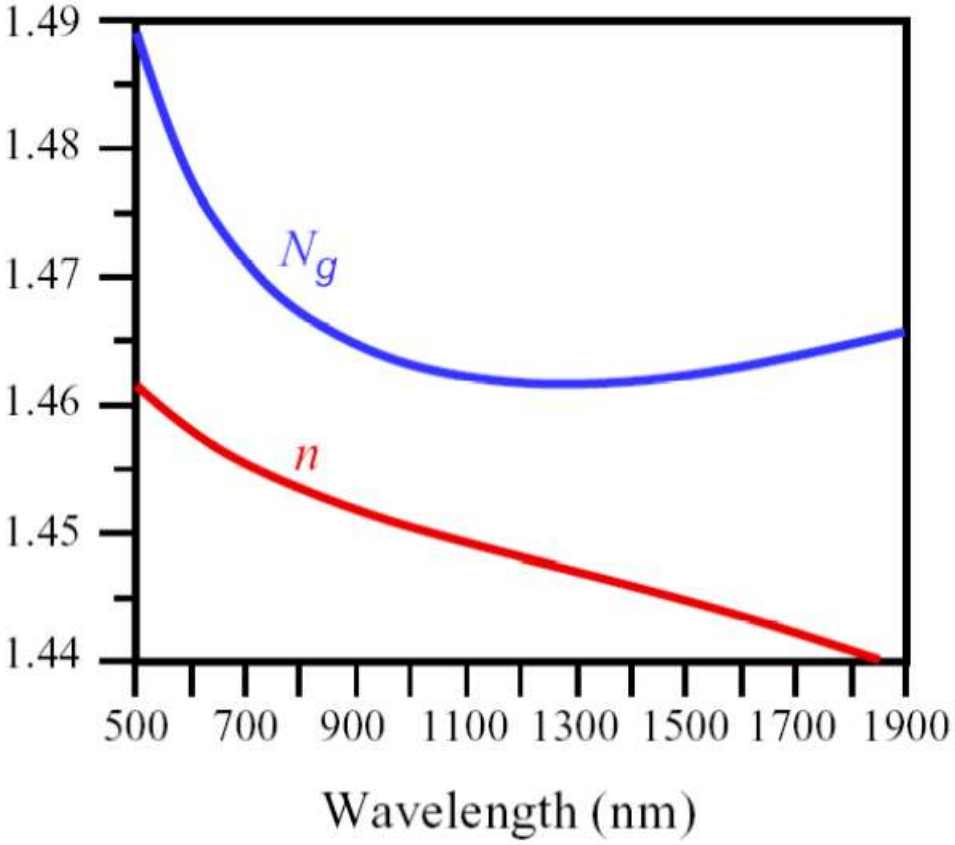
$$ck = n$$

$$\frac{dk}{d\omega} = \frac{cdk}{cd\omega} = \frac{d(ck)}{cd\omega} = \frac{d(n\omega)}{cd\omega} = \frac{1}{c}\left(n + \omega \frac{dn}{d\omega}\right)$$

$$\lambda_0 = \frac{2\pi c}{\omega}$$

$$\omega \frac{dn}{d\omega} = \omega \frac{dn}{d\lambda_0} \frac{d\lambda_0}{d\omega} = -\omega_0 \frac{dn}{d\lambda_0} \frac{2\pi c}{\omega^2} = -\lambda_0 \frac{dn}{d\lambda_0}$$

Group Index of the Medium



For silica, N_g is minimum around 1300 nm, which means that light waves with wavelengths close to 1300 nm travel with the same group velocity and do not experience dispersion.

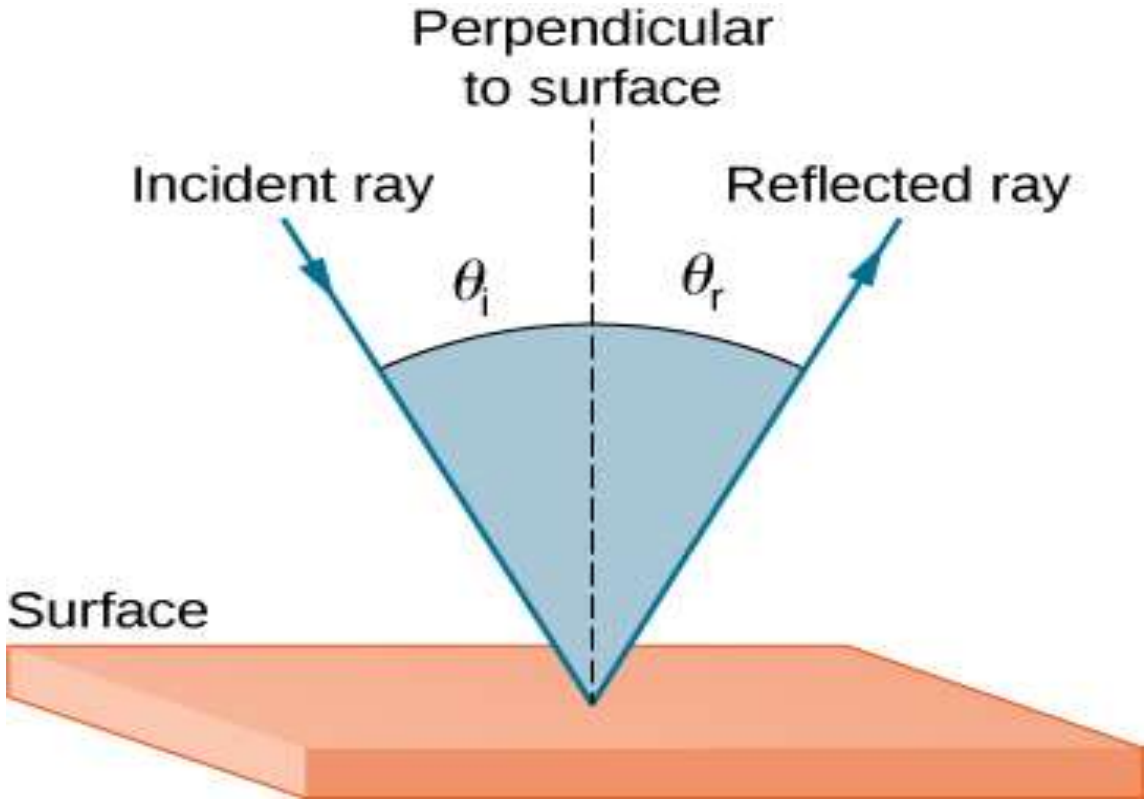
Both n and N_g are functions of the free space wavelength. Such a medium is called a **dispersive medium**. n and N_g of SiO_2 are important parameters for optical fiber design in optical communications.

Ray Transmission Theory

Ray optics: basic laws

Ray optics is based on three laws which describe the propagation of rays:

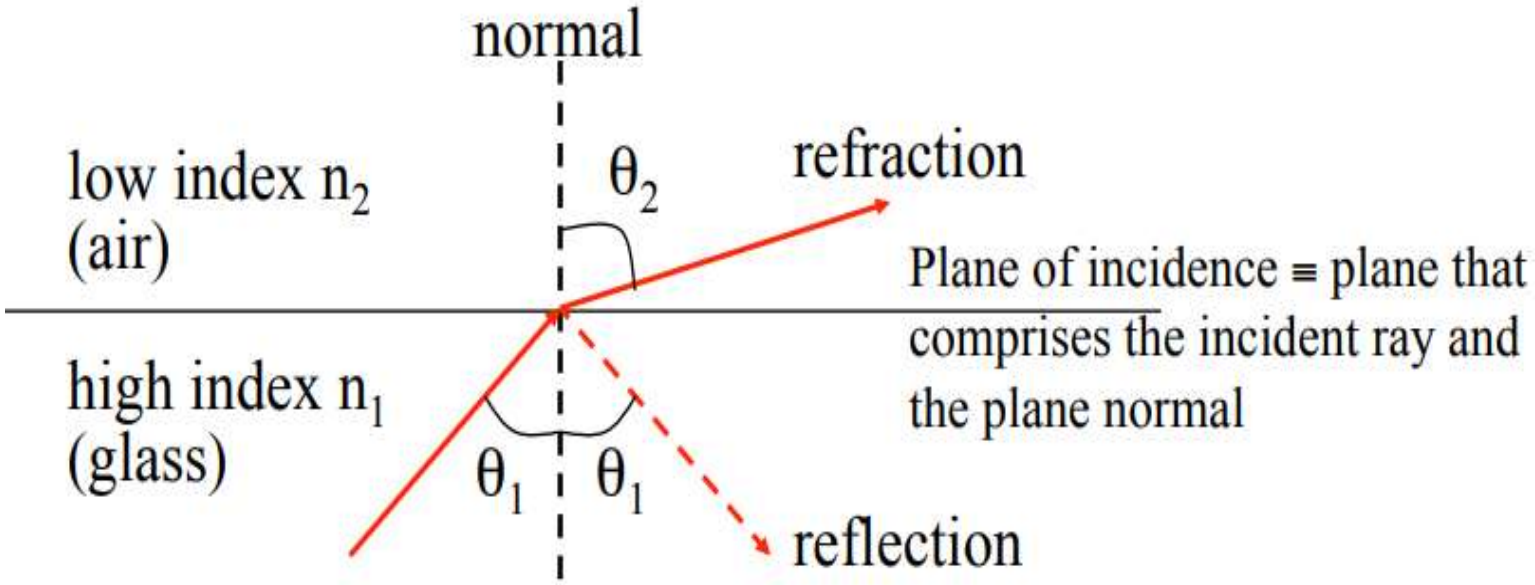
1. Light rays in homogeneous media are straight lines.
2. Law of reflection: Reflection in a mirror or at the boundary between two media of different refractive indices: the reflected ray lies in the plan of incidence, the angle of reflection equals the angle of incidence (i.e. $\theta_r = \theta_i$)



Snell's law

At the boundary between two media of different refractive indices n, the reflected ray lies in the plane of incidence. The angle of refraction θ_t is related to the angle of incidence θ_i by

$n_i \sin\theta_i = n_t \sin\theta_t$.



When a ray is incident on the interface between two dielectrics of different refractive indices (e.g. glass-air), reflection and refraction occur.

The snells law of refraction: $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

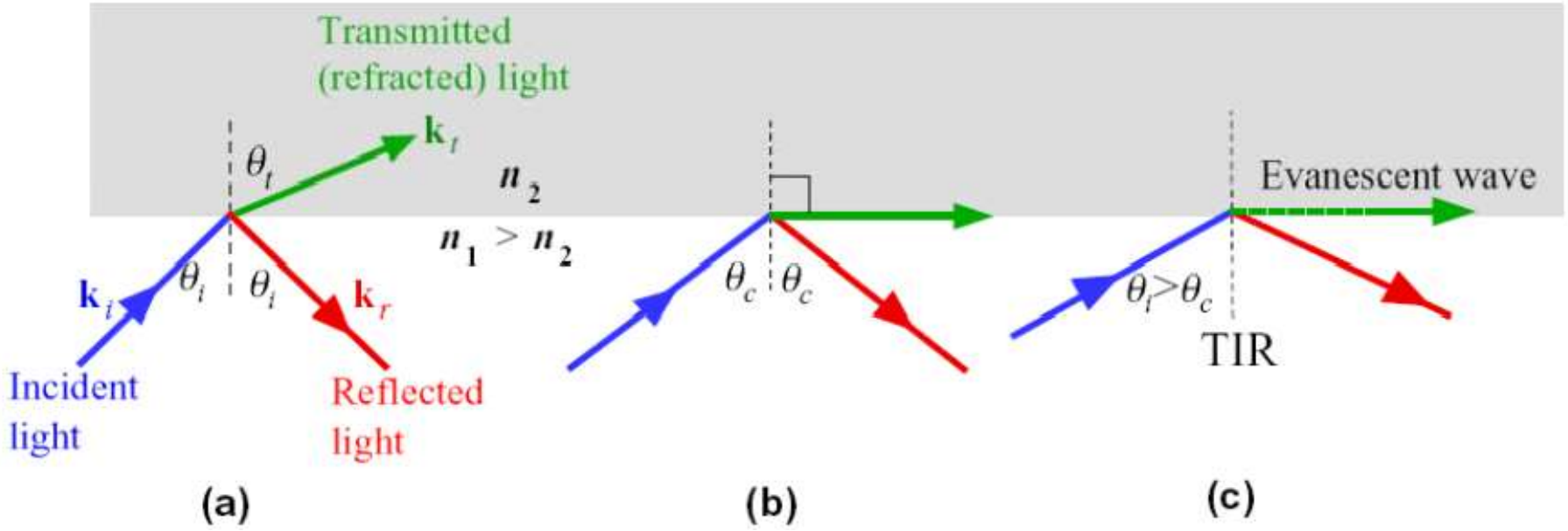
Refraction

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{n_2}{n_1}$$

Total Internal Reflection

When $n_1 > n_2$ and $\theta_t = 90°$, the incidence angle is called the **critical angle** θ_c:

$$\sin\theta_c = \frac{n_2}{n_1}$$



Refraction

A light ray is incident from medium-1 to medium-2. If the refractive indices of medium-1 and medium-2 are 1.5 and 1.36 respectively then determine the angle of refraction for an angle of incidence of 30° .

Solution: From Snell's law: $n_1\sin\theta_1=n_2\sin\theta_2$

$1.5\sin30^0=1.36\sin\theta_2$

$\sin\theta_2=\frac{1.5}{1.36}\sin30^0$

$\sin\theta_2=0.55147$

$\theta_2=33.46^0$

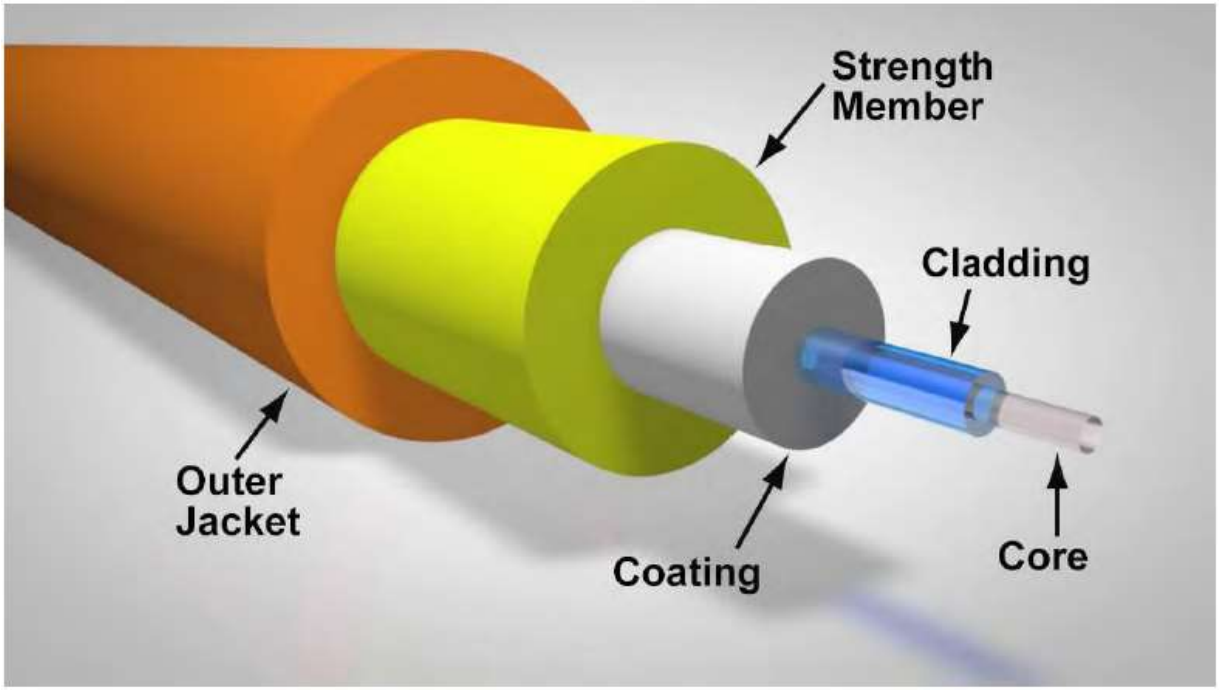
A light ray is incident from glass to air. Calculate the critical angle (θ_c).

Glass,n1=1.5, Air,n2=1

$\sin\theta = n2/n1$

$\Theta = 41.81^\circ$

Optical fiber structures

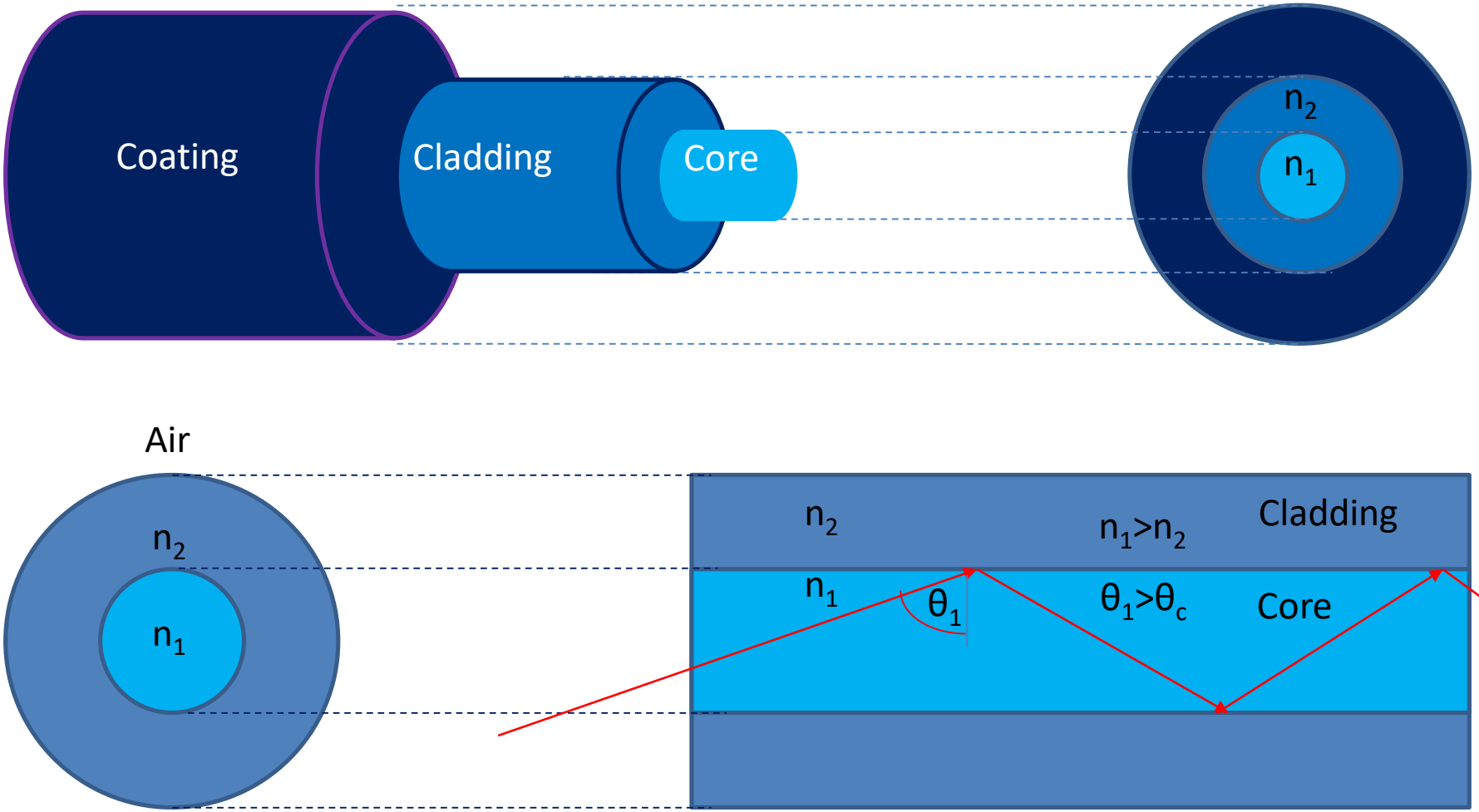


Core: The inner light-carrying layer with a high index of refraction. The core diameter can vary from about 5 µm to 100 µm.

Cladding: The middle layer, which serves to confine the light to the core. It has a lower refraction index than the core.

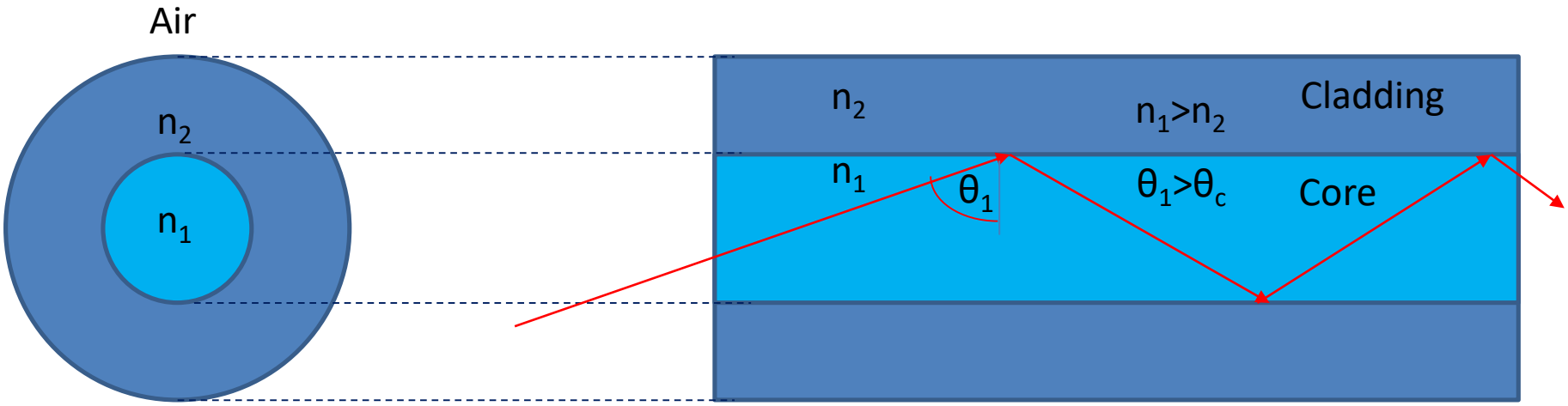
Jacket/Buffer: The outer layer, which serves as a "shock absorber" to protect the core and cladding from damage. The coating usually comprises one or more coats of a plastic material to protect the fiber from the physical environment. Sometimes metallic sheaths are added to the coating for further physical protection. The typical diameter of fiber after coating is 250 µm to 300 µm.

Working Principle of Fiber



- When an ray of light travels from a denser (high index) to a rarer (low index) medium such that the angle of incidence is greater than the critical angle, the ray reflects back into the same medium.
- In the optical fiber the rays undergo repeated total number of reflections until emerges out of the other end of the fiber, even if fiber is bend.

Light ray guiding condition



The refractive index of a core is $n_1 = 1.46$, and cladding $n_2 = 1.44$. Under what conditions will light be trapped inside the core?

Solution:

$\theta_c = \sin^{-1}(n_2/n_1) = \sin^{-1}(1.44/1.46) = 80.5^{\circ}$

If $\theta_i > \theta_c$, the light will be trapped inside the core.

Silica optical fibers

Both the core and the cladding are made from a type of glass known as silica (SiO_2) which is almost transparent in the visible and near-IR.

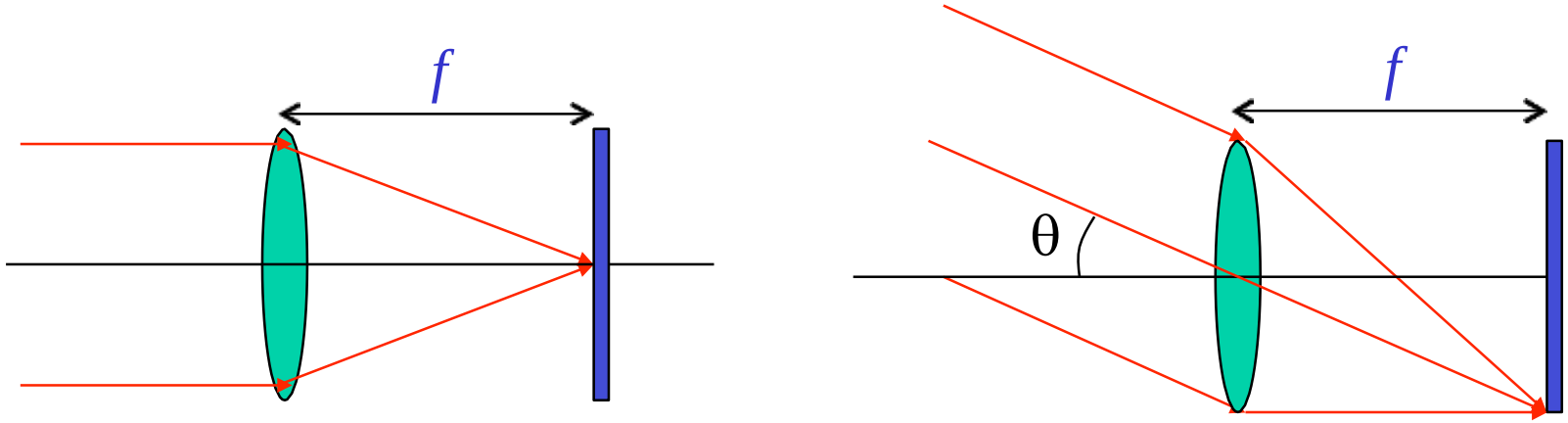
The higher core refractive index (~ 0.3% higher) is typically obtained by doping the silica core with germanium dioxide (GeO_2).

In Lab1, we shall learn how to strip off the jacket to expose the cladding. This is necessary in order to “cleave” the fiber for a smooth end face for light coupling.

In Lab 1, we should be able to see the step boundary between the core and the cladding, by end-illuminating the fiber and imaging the output-end cross-section using a microscope.

Numerical aperture

An important characteristic of an optic system is its ability to collect light incident over a wide range of angles.



The numerical aperture (NA) is defined as:

$$NA = n_o \sin\theta$$

where n_o is the refractive index of the medium between the lens and the image plane (e.g. a photodetector) and θ is the maximum acceptance angle.

Numerical aperture

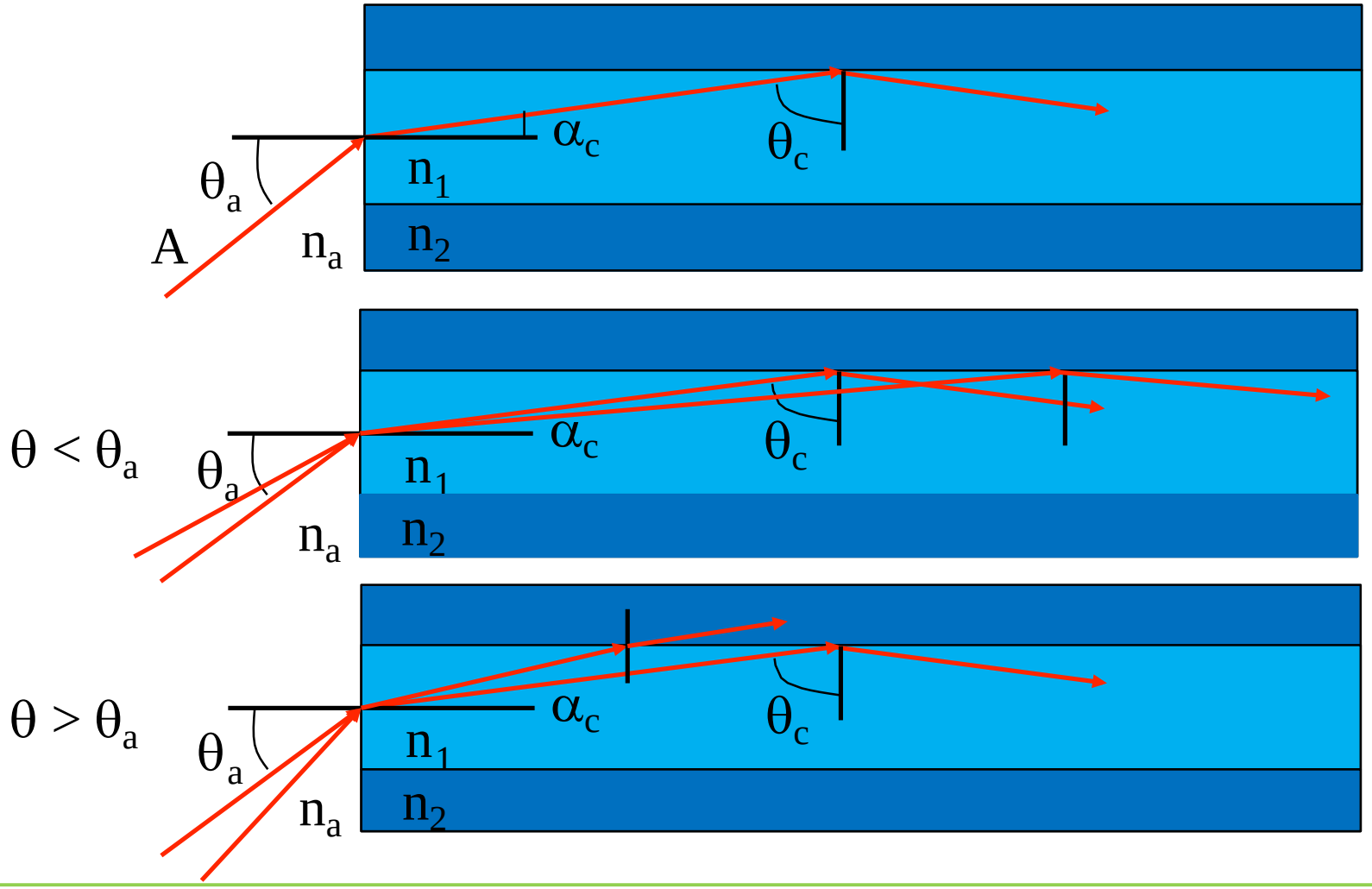
- The definition of numerical aperture applies to *all light-collecting systems*, including optical fibers.
- e.g. Light rays incident at angles *outside* the collection cone for a fiber will *not* propagate along the fiber (*instead will attenuate rapidly*).
- The numerical aperture is often measured in air, $n_o = 1$

$$NA = \sin\theta$$

- A *low* NA indicates a *small* acceptance angle.
- Light coupling to a low-NA optical system (e.g. fiber) is more difficult (*alignment is more sensitive*) and less efficient (*some of the rays are outside the acceptance angle*) than is coupling to a high-NA optical system.

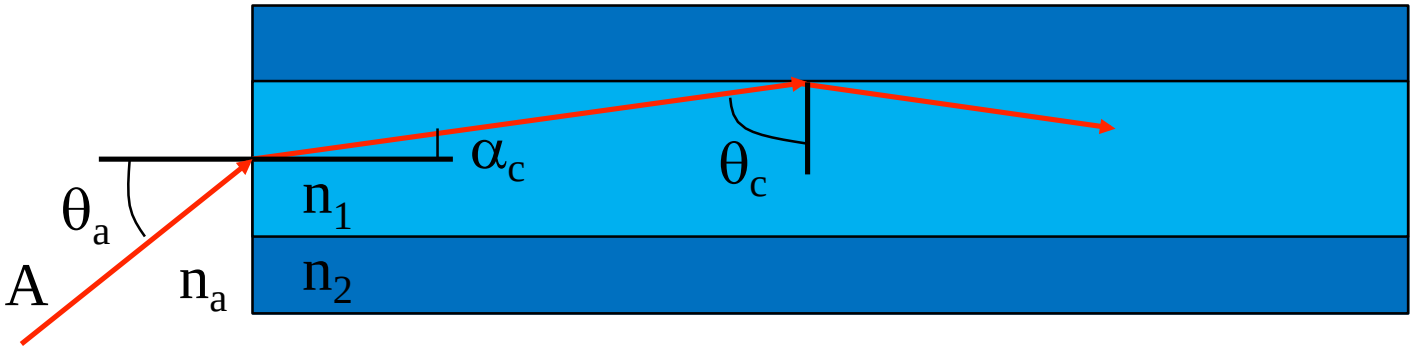
Acceptance angle

Only rays with a sufficiently shallow grazing angle (i.e. with an angle to the normal greater than θ_c) at the core-cladding interface are transmitted by total internal reflection.



Light rays will be confined inside the fiber core if it is input-coupled at the fiber core end-face within the acceptance angle θ_a.

Acceptance angle



e.g. What is the fiber acceptance angle when n_1=1.46 and $n_2 = 1.44$?

$\theta_c = \sin^{-1}(n_2/n_1) = 80.5^o \Rightarrow \alpha_c = 90^o - \theta_c = 9.5^o$

using, $\sin\theta_a = n_1 \sin\alpha_c$ (taking $n_a = 1$)

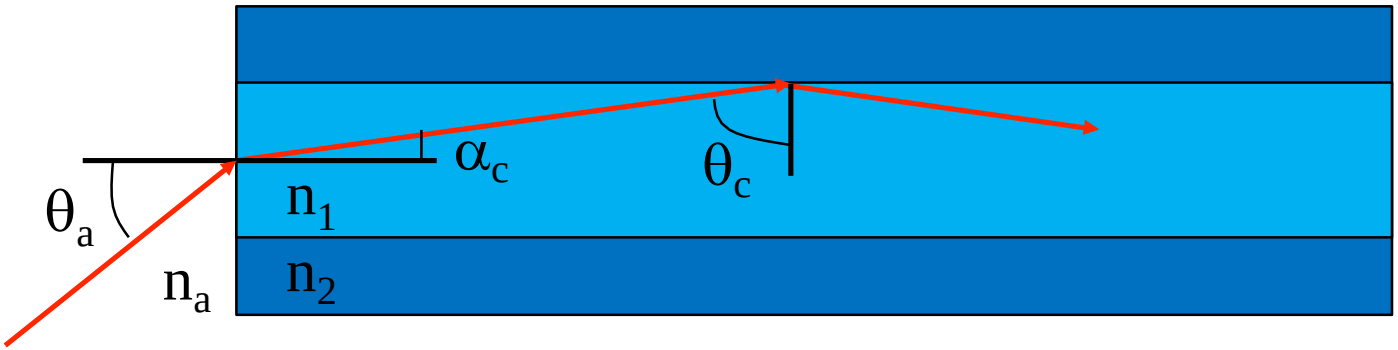
$\theta_a = \sin^{-1}(n_1 \sin\alpha_c) = \sin^{-1}(1.46 \sin 9.5^o) \sim 14^o$

$\Rightarrow$the acceptance angle $\theta_a \sim 14^o$

Fiber numerical aperture

In fiber optics, we describe the fiber acceptance angle using **Numerical Aperture (NA):**

$$NA = n_a \sin\theta_a = \sin\theta_a = (n_1^2 - n_2^2)^{1/2}$$



•we consider the refraction at the air-core interface using Snell's law:

At θ_a: $n_a \sin\theta_a = n_1 \sin\alpha_c$

launching the light from air ($n_a \sim 1$):

$$\sin\theta_a = n_1 \sin\alpha_c$$
$$= n_1 \cos\theta_c$$
$$= n_1 (1 - \sin^2\theta_c)^{1/2}$$
$$= n_1 (1 - n_2^2/n_1^2)^{1/2}$$
$$= (n_1^2 - n_2^2)$$

e.g. What is the fiber numerical aperture when $n_1 = 1.46$ and $n_2 = 1.44$?

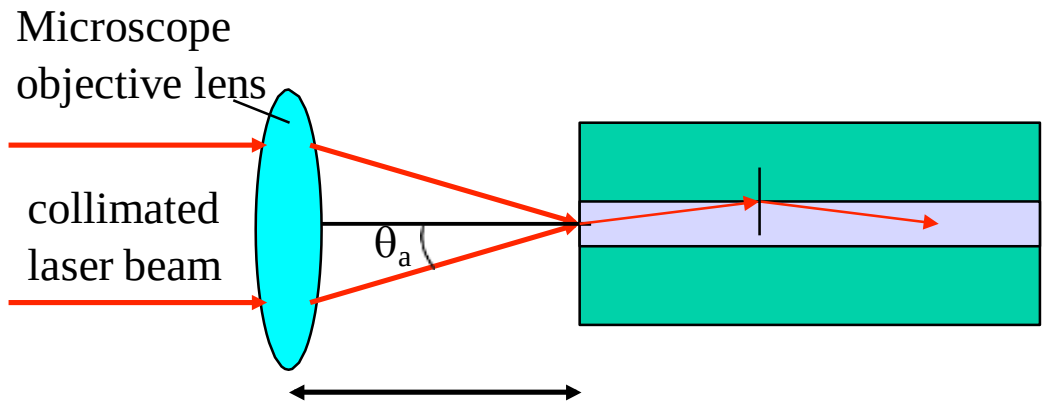
$$NA = \sin\theta_a = (1.46^2 - 1.44^2)^{1/2}$$
$$= 0.24$$

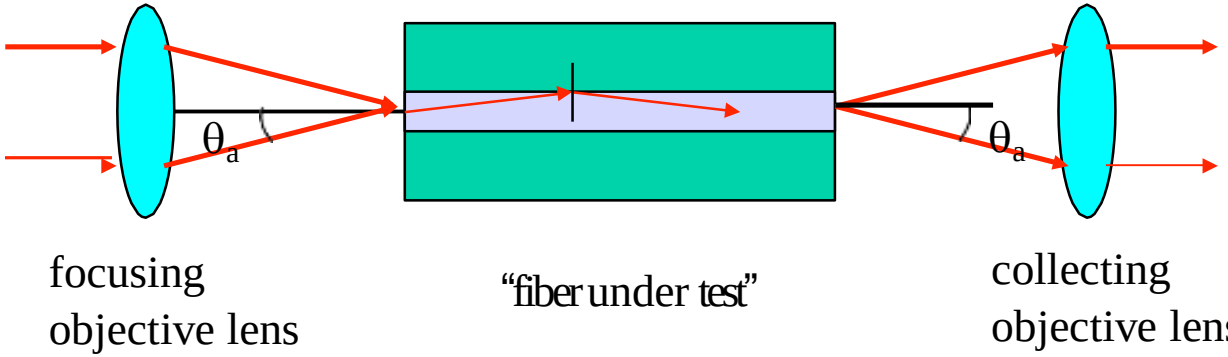
It is a common practice to define a relative refractive index Δ as:

$$\Delta = (n_1 - n_2) / n_1$$

$NA = n_1(2\Delta)^{1/2}$ if $n_1 \sim n_2$

Lens coupling to fiber end faces





•By measuring the output couple ray cone angle, we can measure the fiber acceptance angle.

Large-NA fibers?

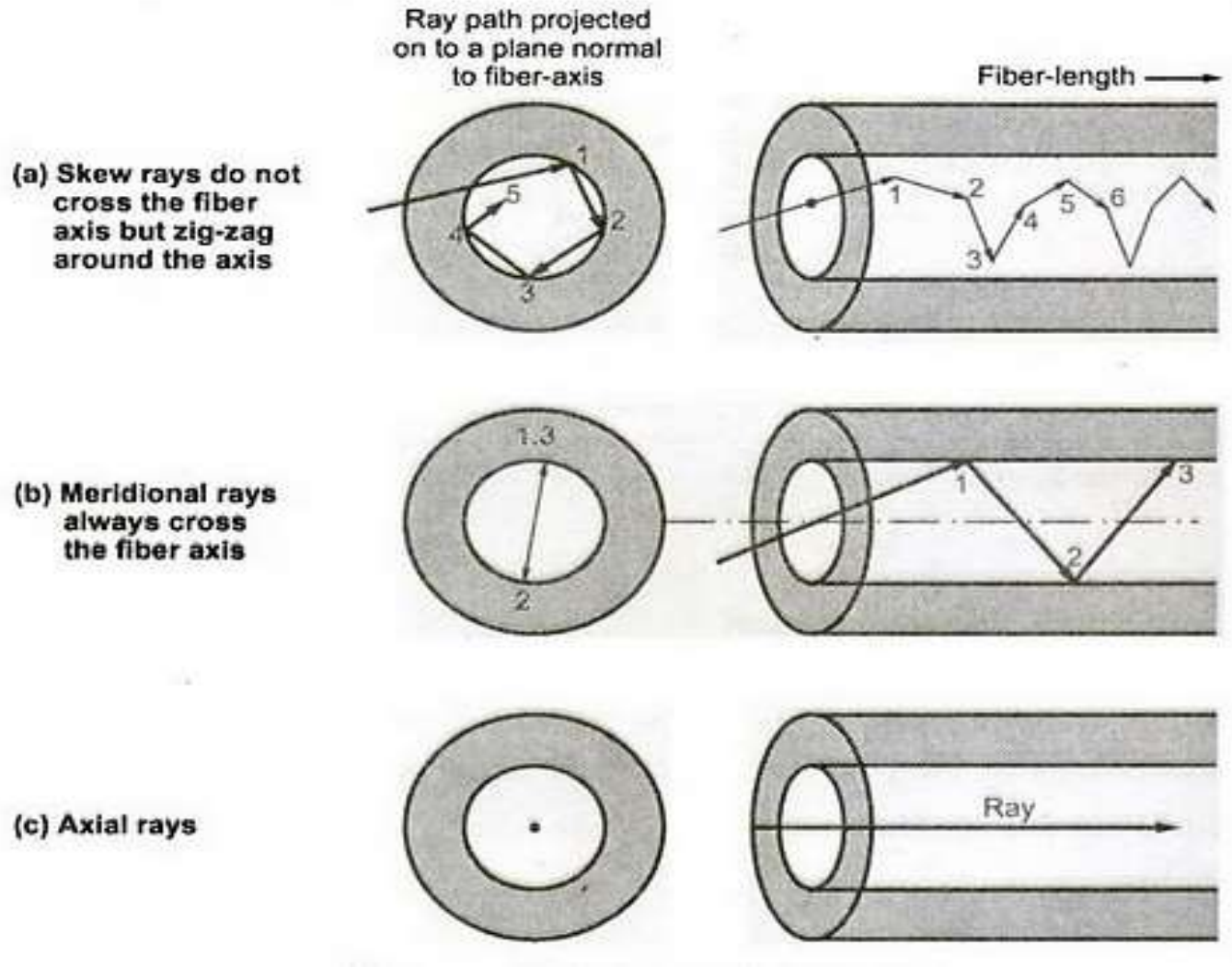
- Developing ways for fiber to collect light efficiently was an important early step in developing practical fiber optic communications (particularly in the 1970s).
- It seems logical to have optical fibers with NA as large as possible … with as large Δ as possible … in order to couple maximum amount of light into the fiber.
- Soon, we will find out that such large-NA fibers tend to be "multimode" and are unsuitable for high-speed communications because of a limitation known as modal dispersion.
- Relatively small-NA fibers are therefore used for high-speed optical communication systems.

Typical fiber NA

- Silica fibers for long-haul transmission are designed to have numerical apertures from about 0.1 to 0.3.
- The low NA makes coupling efficiency tend to be poor, but turns out to improve the fiber's bandwidth! (details later)
- Plastic, rather than glass, fibers are available for short-haul communications (e.g. within an automobile). These fibers are restricted to short lengths because of the relatively high attenuation in plastic materials.
- Plastic optical fibers (POFs) are designed to have high numerical apertures (typically, 0.4 – 0.5) to improve coupling efficiency, and so partially offset the high propagation losses and also enable alignment tolerance.

Types of Ray

Ray is an ideal narrow beam of light that is used to model the propagation of light through an optical system.



Limitation of ray optics

- For smaller fiber diameters that are only few times of the wavelength, geometrical optics approach becomes inadequate.This is because ray optics only describes the direction a plane wave component takes in the fiber, but does not take into account interference among such components.

- When interference phenomena are considered it is found that only rays with certain discrete characteristics propagate in the fiber core.

- Thus the fiber will only support a discrete number of guided modes.

- This becomes critical in small core diameter fibers which only support one (singlemode) or a few modes (multimode). Electromagnetic theory must be applied in this case.